

Mission 1: AtmosphereWatch CubeSat

Mission idea

Students use the CubeSat to collect **temperature, pressure, and humidity** data at different locations or elevations to model how atmospheric conditions change and how those changes can be monitored like a weather or Earth-observing satellite.

Driving question

How can a CubeSat monitor changing atmospheric conditions to help people better understand weather and environmental systems?

Mission objectives

- Measure temperature, humidity, and pressure over time
- Compare data across locations, heights, or times of day
- Identify environmental patterns and possible hazard indicators
- Evaluate how well the CubeSat design protects and supports reliable sensing

Example student activities

- Collect indoor vs. outdoor data
- Take data at ground level vs. upper floor/stairwell
- Monitor a sealed container vs. open air
- Compare sunny vs. shaded conditions
- Graph pressure, humidity, and temperature trends

NGSS alignment

- **MS-ESS3-2:** Use environmental data to discuss hazard monitoring
- **MS-ESS3-3:** Design a method to monitor environmental conditions

- **MS-ESS3-5:** Discuss temperature trends and environmental change
- **MS-PS1-4:** Relate heating/cooling to temperature changes
- **MS-ETS1-1 to 1-4:** Define the problem, test designs, analyze data, improve the CubeSat